Institutional and Water-Governance Response

Institutional and Water-Governance Response

History of Chile's Water Code Prior to 1981, water in Chile was managed primarily through state-administered concessions. The 1969 Water Code strengthened government control by treating water as a public resource and limiting private transfers of water rights (Ríos & Quiroz, 1995). The 1981 Water Code fundamentally changed this approach by separating water rights from land ownership, establishing private property rights over water, and allowing rights to be freely traded. The reform aimed to improve allocative efficiency, encourage investment, and reduce state involvement in water management by relying on market mechanisms to direct water toward its highest-value use (Bauer, 1997; Prieto et al., 2019). While proponents argued that water markets improved flexibility and economic efficiency, critics contended that high transaction costs, weak oversight, and unequal access limited market performance and failed to address environmental and social externalities (Bauer, 1997; Boelens, 2009).

A further challenge emerged through the over-allocation of water rights, often referred to as "paper water". Under the 1981 framework, the DGA had limited authority to deny applications when rights appeared legally available, resulting in the issuance of water entitlements that exceeded physically available resources in several basins and aquifers. Consequently, the volume of legally granted rights often surpassed actual hydrological availability, creating structural imbalances between paper entitlements and real water supplies (Prieto et al., 2019).

Push Towards Reform Three pressures converged to force institutional change. First, the 2010–2022 megadrought turned chronic over-allocation into acute shortage across the mining north and agricultural centre. Second, the state proved unable to reallocate: constitutional protection of private rights left the DGA able to act only under formally declared scarcity, while basins were already legally exhausted. Third, the desiccation of the Loa wetlands and the dispossession of Atacameño and peasant communities, alongside rural drinking-water crises, exposed water commodification's social costs (Prieto et al., 2019; Suárez Delucchi, 2024).

In response, the DGA sought to recover authority. The 2005 amendment introduced the *patente por no uso* — an escalating fee on idle rights — and required applicants to justify intended use (Donoso, 2015). The decisive shift came with the 2022 Water Code Reform; it ended perpetual ownership, converting rights into 30-year revocable concessions, and established a statutory priority cascade placing human consumption and ecosystem flow above productive use, with new powers to prohibit rights over glaciers, create reserves, and expropriate for domestic supply (Macpherson et al., 2023). The DGA's existing instruments — aquifer restriction and prohibition zones, river exhaustion declarations, and scarcity decrees — became the operational map of where rights now exceed supply (Appendix 2, panel 1).

Impact on the mining sector The mining sector became a central focus of these evolving governance arrangements. Although Chile did not enact a law explicitly requiring mines to use desalinated water, increasing restrictions on freshwater extraction, stricter environmental permitting requirements (RCA), enhanced DGA oversight, and growing social opposition to groundwater use collectively encouraged mining companies to invest in desalination infrastructure. The experience of the Copiapó and Loa basins, where water rights overallocation and groundwater depletion generated significant conflicts among mining companies, agricultural users, and local communities, contributed to a broader policy shift toward desalination as a strategic source of water supply for northern Chilean mining operations (Prieto et al., 2019). The spatial overlap shows that 13 of 19 major copper mines operate inside or within 10 km of a legally restricted or prohibited aquifer or an exhausted river basin (Appendix 1). The firm response to move to desalination is a

rational one to ensure sustainability of future activities. Reflecting this, seawater rose from 11.6% to 34.1% of mining freshwater extraction over 2014–2024, a transition concentrated in the northern mining regions.

Cerro Colorado (BHP, Tarapacá) is the warning where over-extraction from the Lagunillas aquifer damaged protected wetlands and the mine eventually closed in 2023. The institutional environment no longer merely permits adaptation; it imposes the cost of inaction on firms still dependent on continental freshwater.

Data This section draws on three public data sources. Spatial information on Chile's water-governance constraints is obtained from the Dirección General de Aguas (DGA), specifically its Catastro Público de Aguas and associated Mapoteca Digital (updated November 2025), which provide georeferenced layers for aquifer restriction and prohibition zones, exhausted-river declarations, drought-scarcity decrees, protected wetlands and glaciers, and Article 5 reserved flows. These layers are combined with the georeferenced locations of Chile's nineteen largest copper operations, drawn from the project mine registry, to situate mines relative to legally constrained water bodies. Sector-level water use comes from the Comisión Chilena del Cobre (COCHILCO), whose annual report *Consumo de Agua en la Minería del Cobre* (2024) records mining withdrawals by source — surface, groundwater, third-party, and seawater — and underpins the 2014–2024 figures reported above.

Institutional and Water-Governance Response

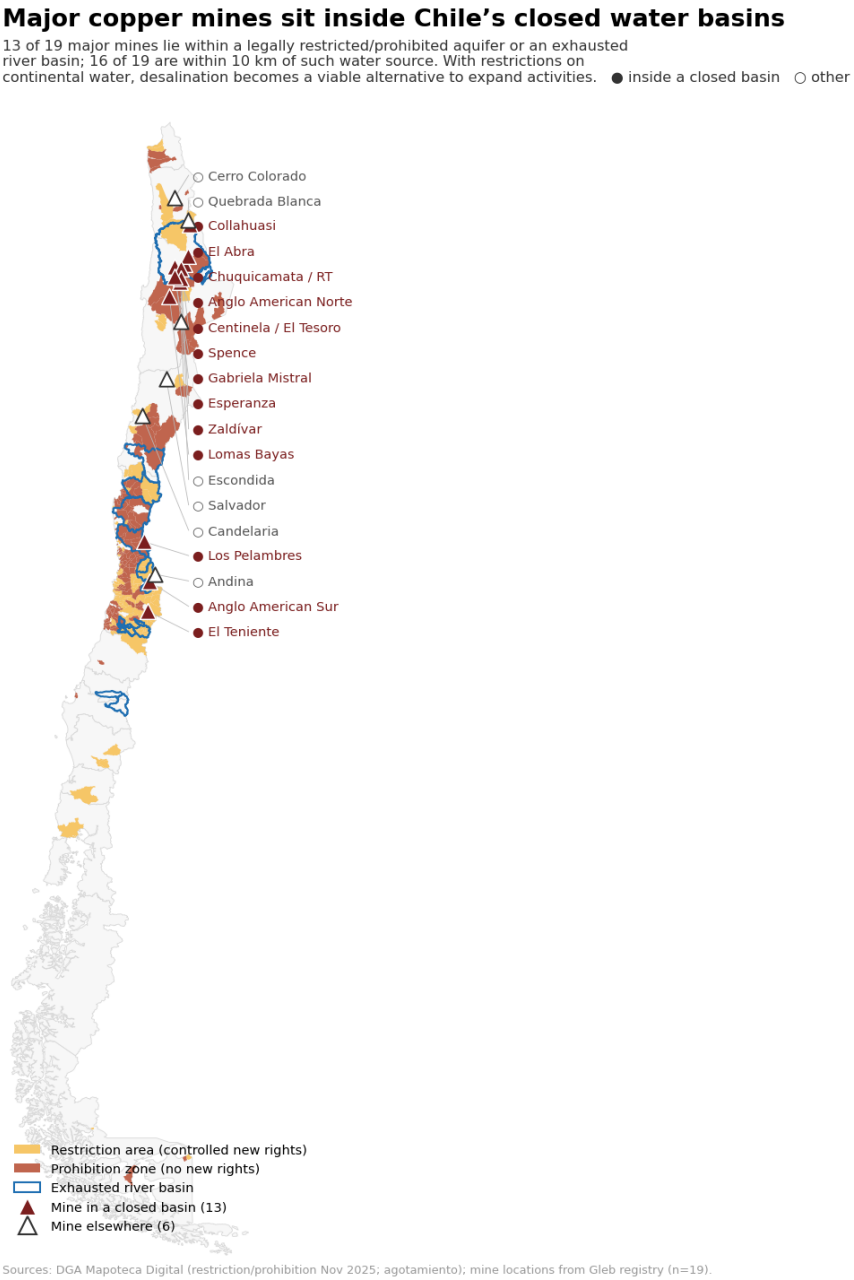


Figure 1: Major copper mines sit inside Chile’s closed water basins

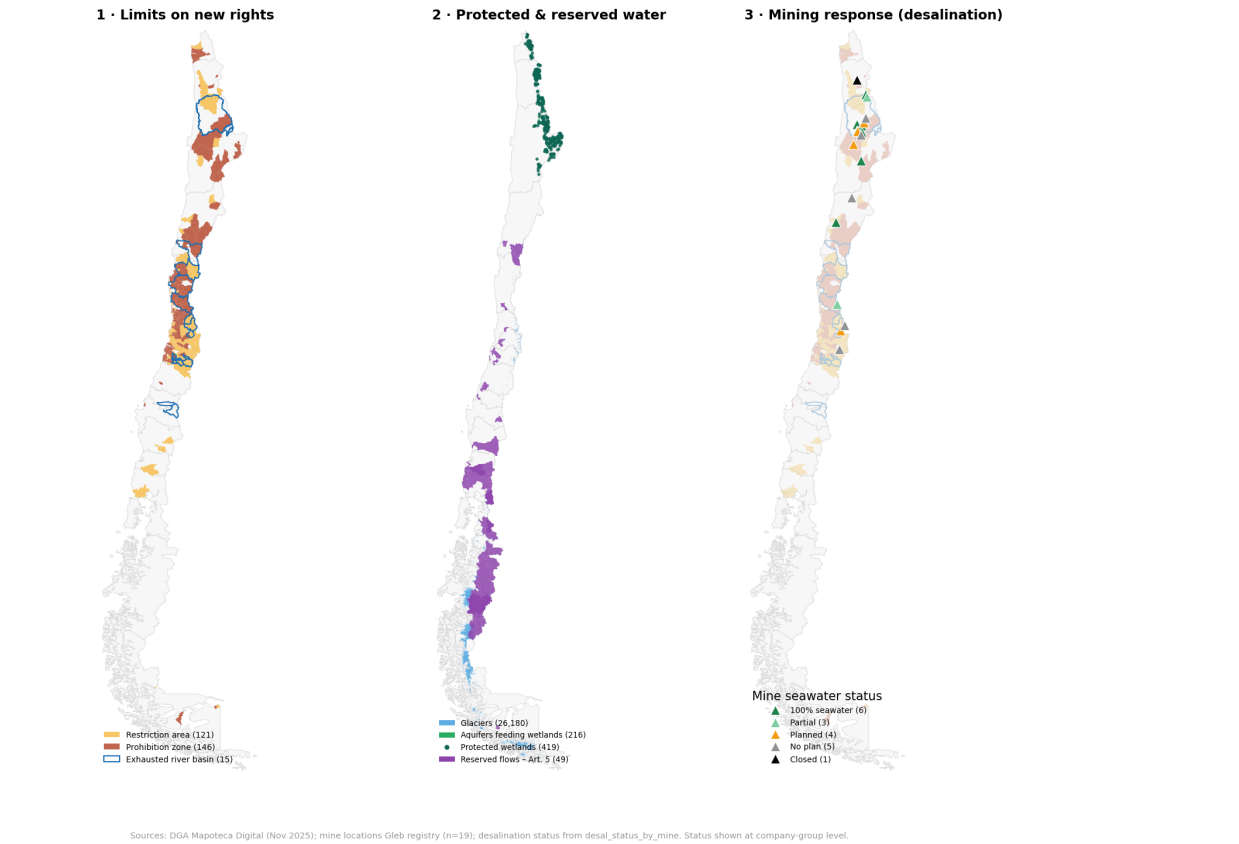


Figure 2: Chile's water-governance constraints and the mining response